



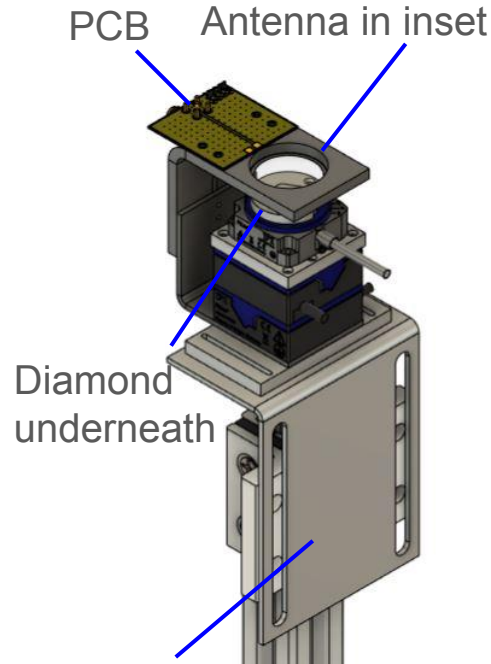
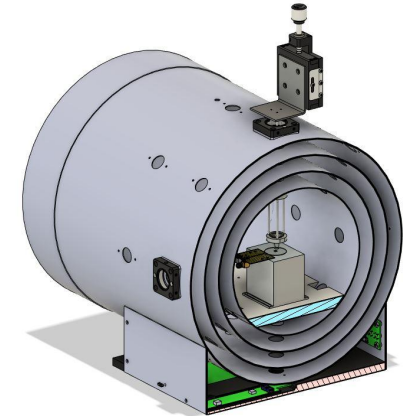
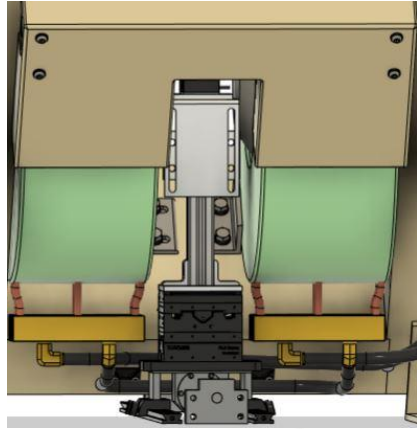
BERKELEY LAB

Coherent Control of Nitrogen Vacancy Centers in Diamond using Broadband Micro-coil Antennas

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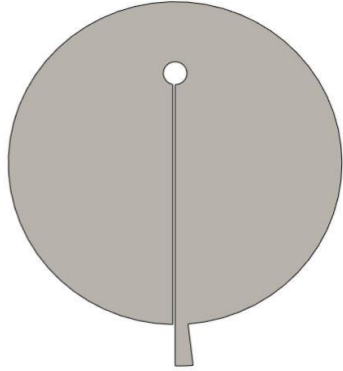
Motivation: Quantum Microscope, Magnetometer



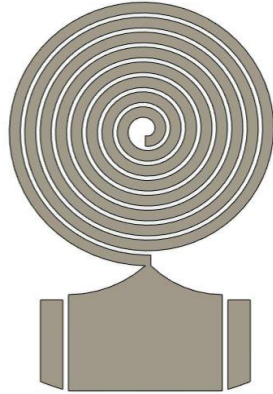
Custom sample stage with XYZ control

- Quantum microscope: NMR with newly small sample volumes for chemical/biological imaging (DOE funded) - built and taking data
- Magnetometer: newly sensitive, portable design for unexploded ordnance (UXO) detection, Earth magnetic field variation, etc (SERDP funded)
- To do these, need to control $\sim 10^{13}$ spins ([1]) with a small, efficient antenna
- Low power necessary (don't want to kill live cells!) - max of ~ 1 mW
- Goal: high contrast at low power, high resolution on small spot, extremely uniform magnetic field, compact

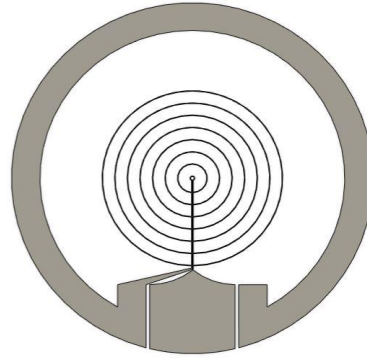
Compact antenna designs all fit on 1" glass cover slip.



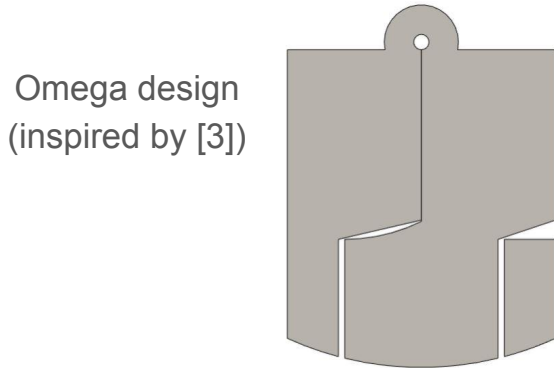
Single ring design
(inspired by [2])



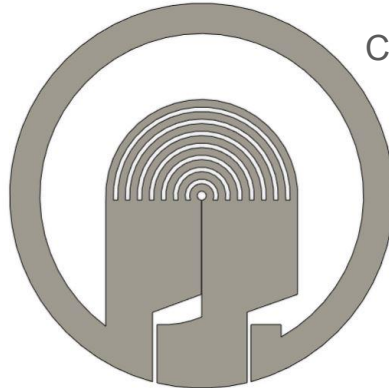
Spiral design
(inspired by [4])



Concentric design
(new in literature)



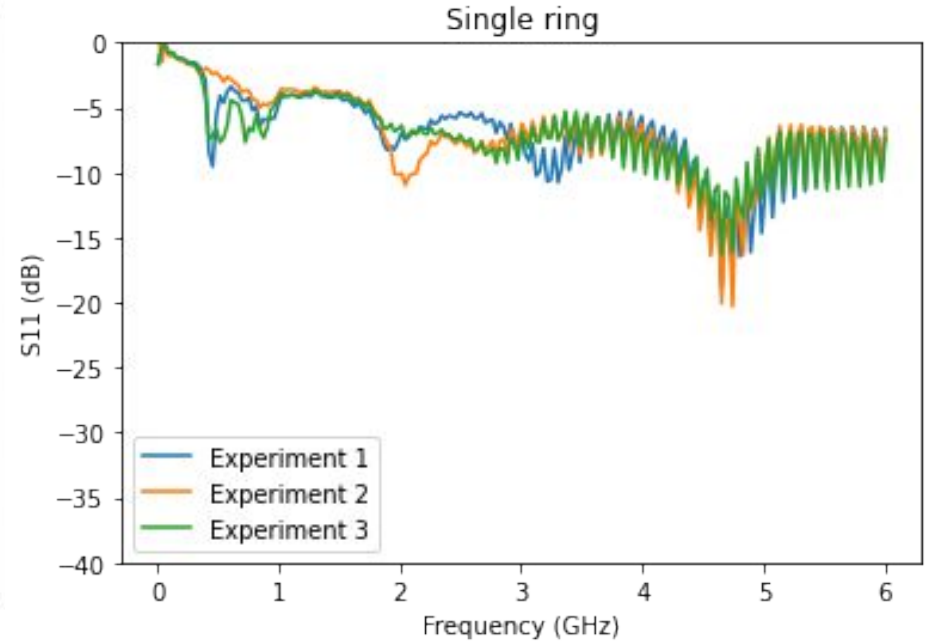
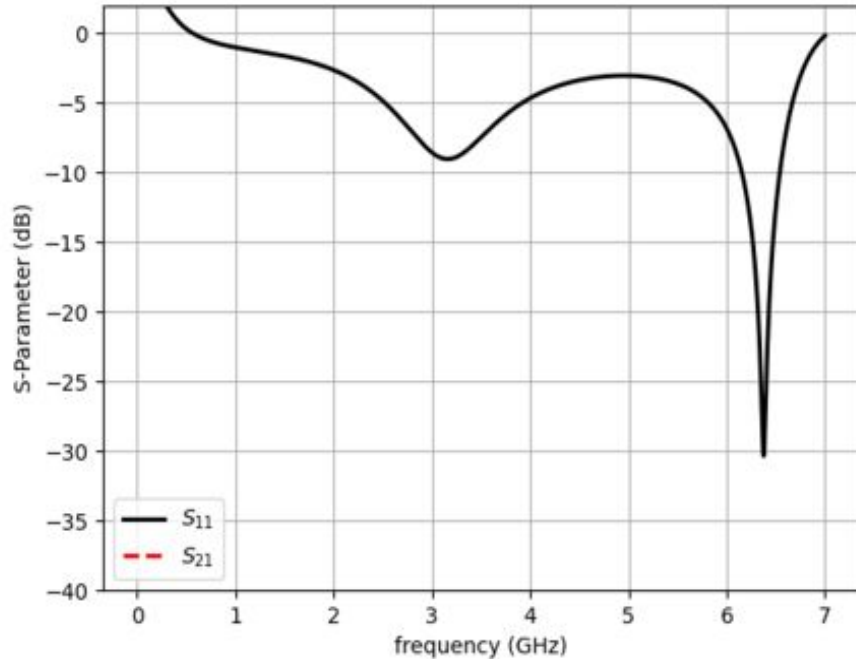
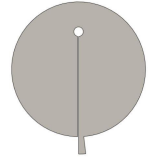
Omega design
(inspired by [3])



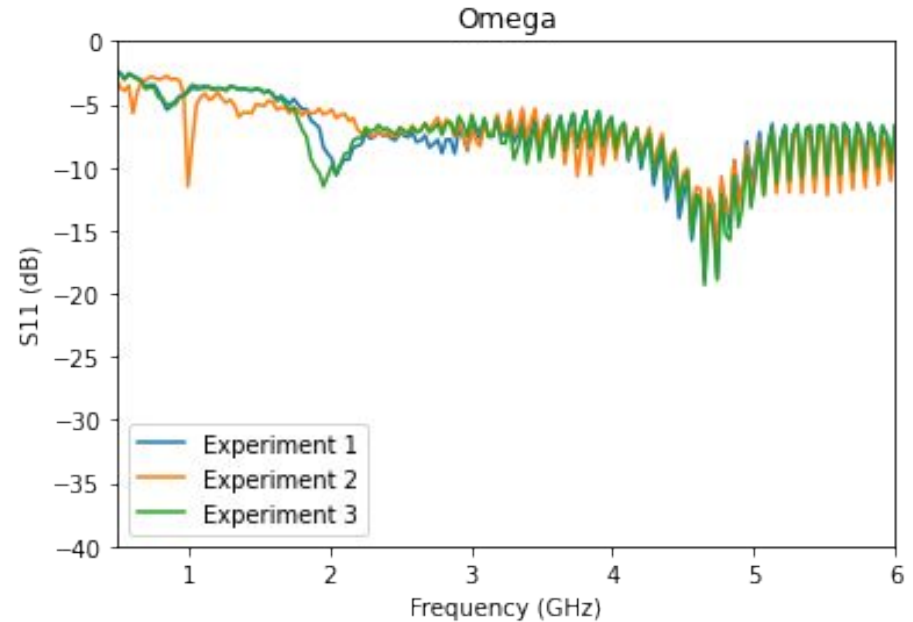
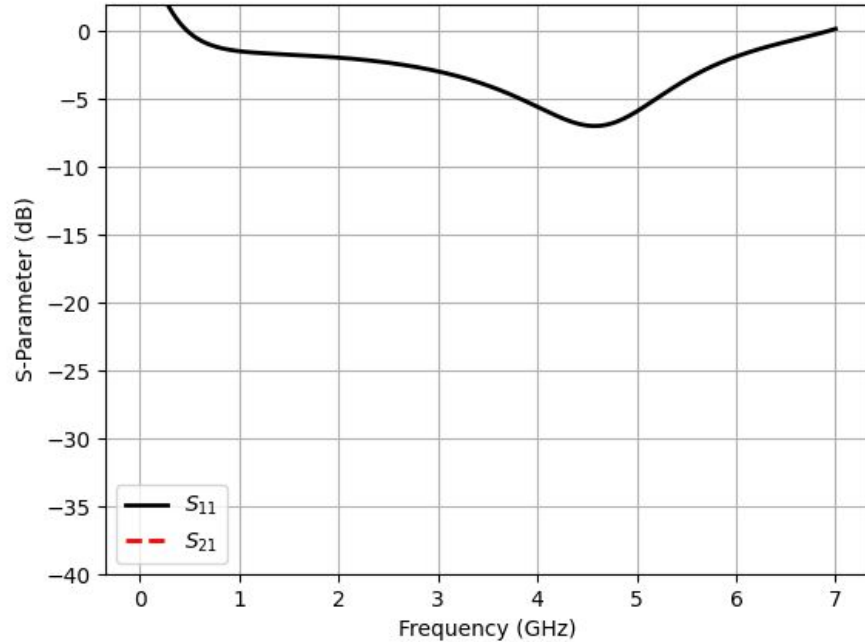
Concentric omega
design (new in
literature)

- Spiral shapes inspired by NMR designs, concentric and spiral more broadband resonance
- Omega shape intended to have deep resonance
- Best sensitivity to magnetic field/chemical species

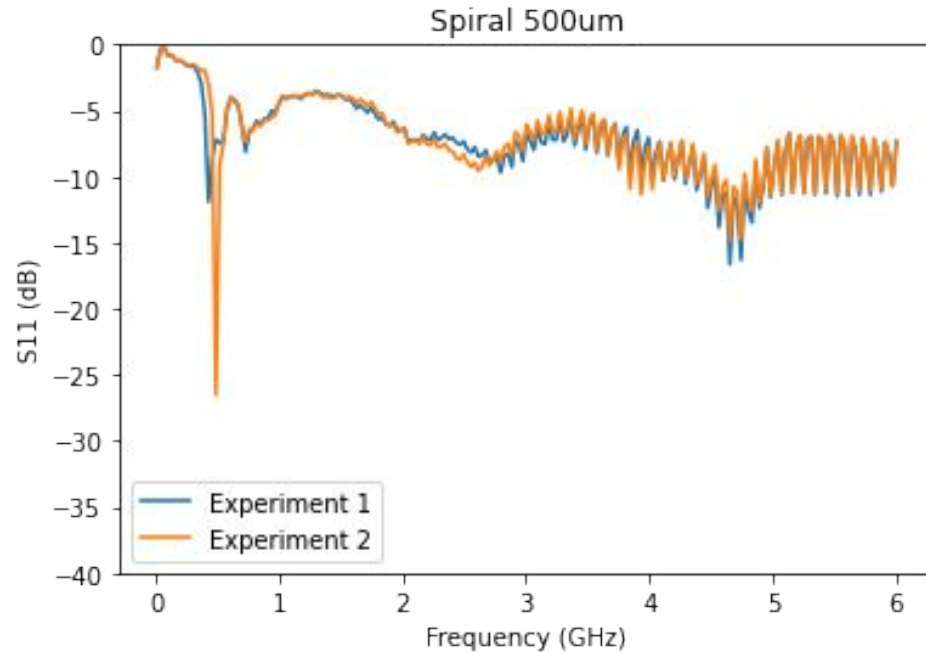
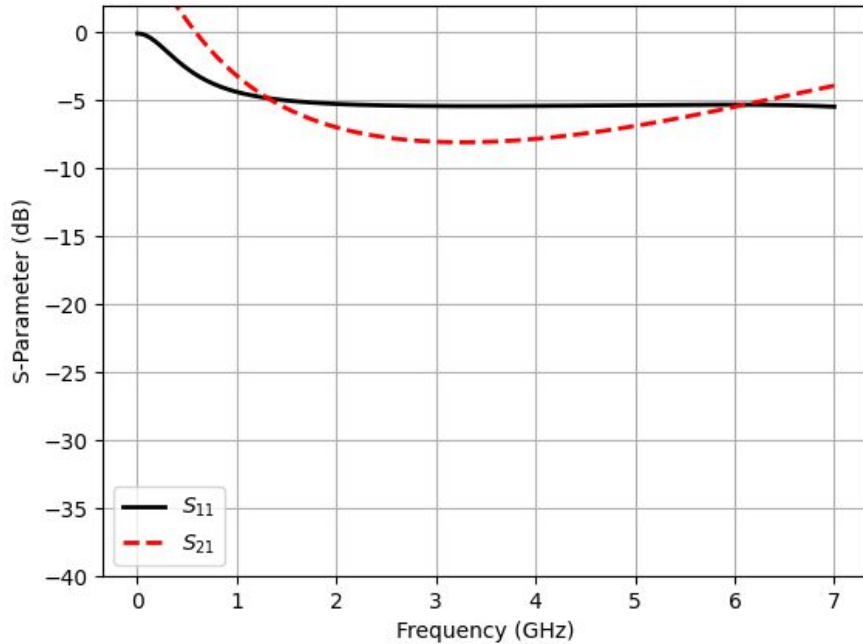
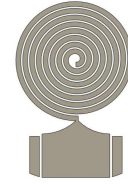
Simulation & characterization: Single ring antenna



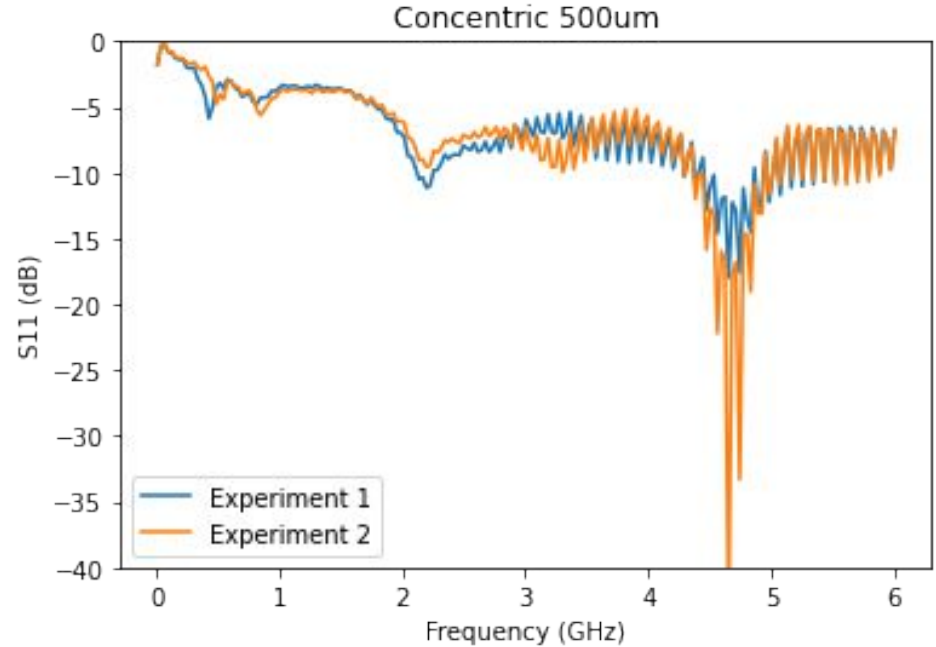
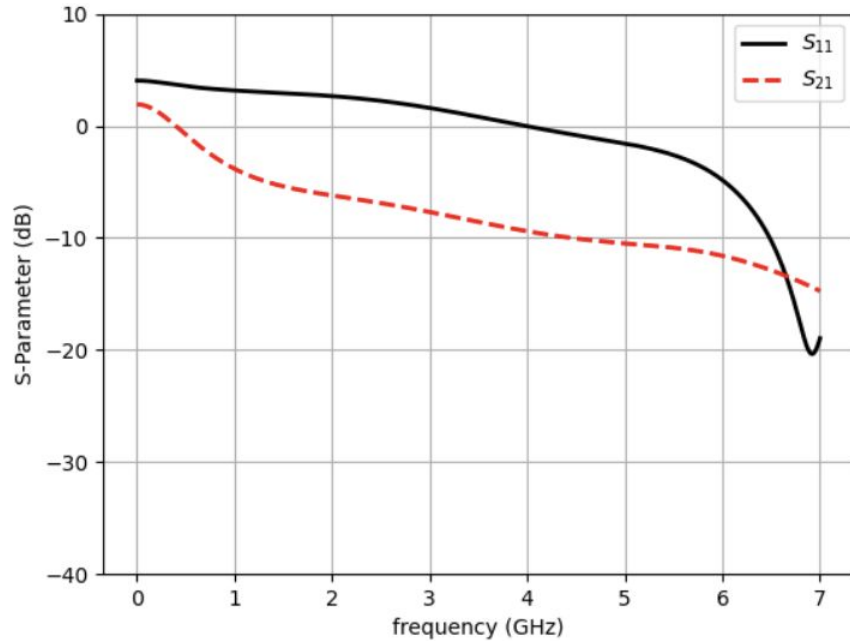
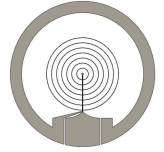
Simulation & characterization: Omega antenna



Simulation & characterization: Spiral antenna

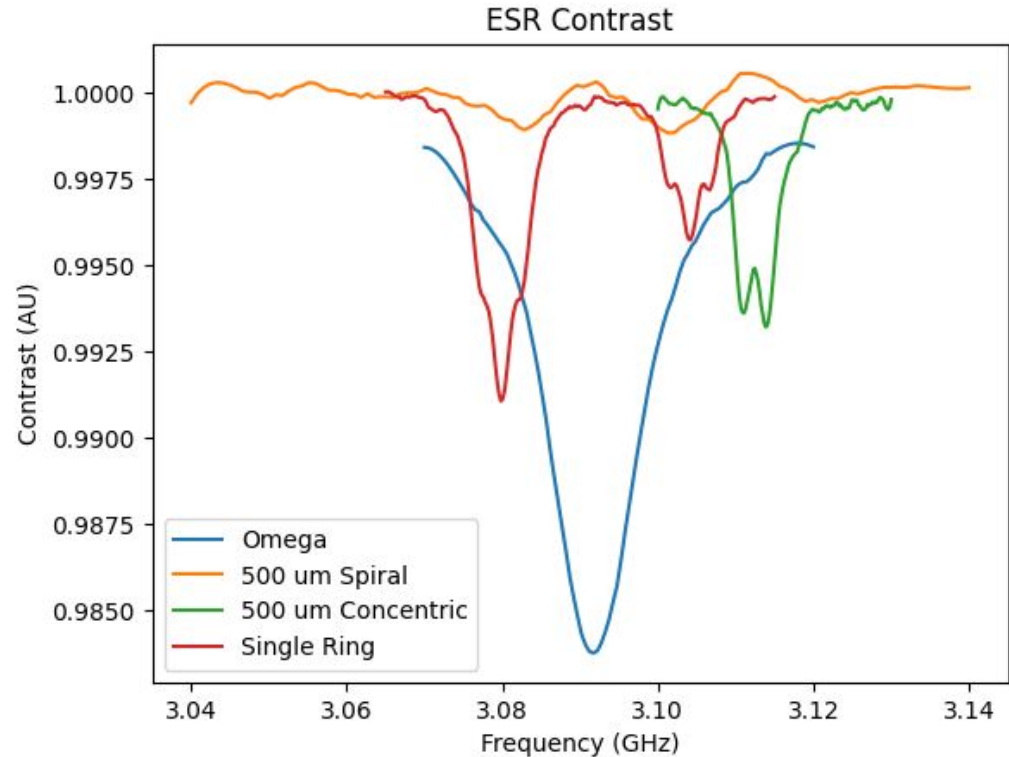


Simulation & characterization: Concentric antenna



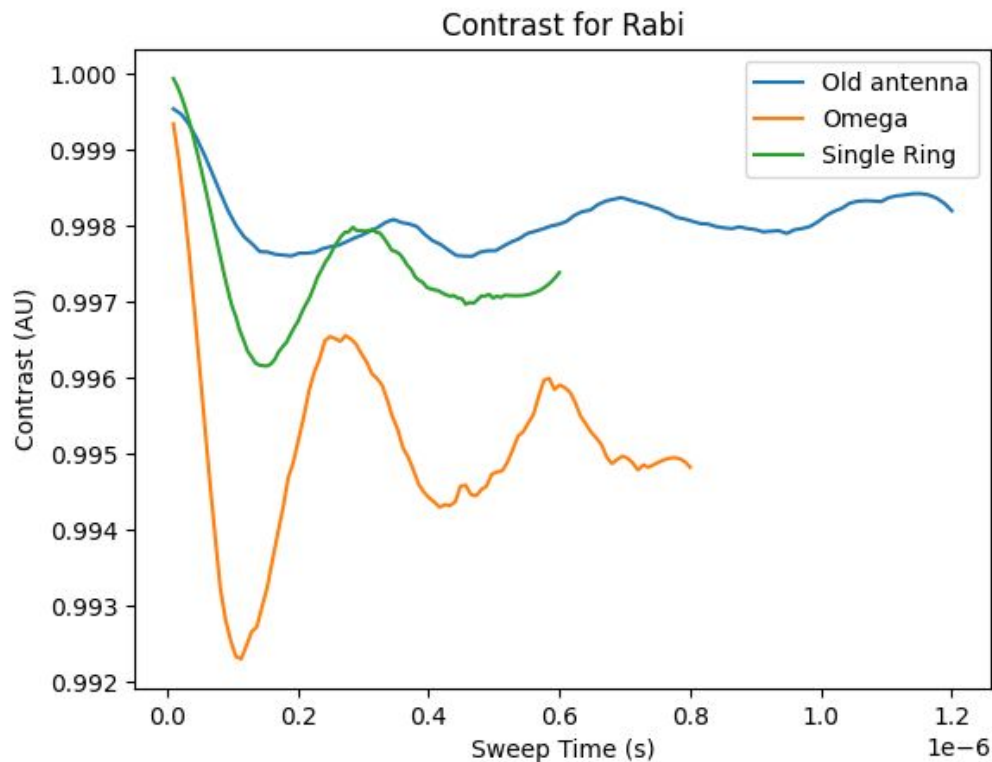
ESR data shows concentric, single ring, omega good performance.

- Number of peaks dependent on sample alignment, independent of antenna performance
- Spiral antenna performed poorly, comparably to old loop antenna
- Omega antenna adapted from [3] performed best
- Concentric antenna new, quality design comparable to single loop



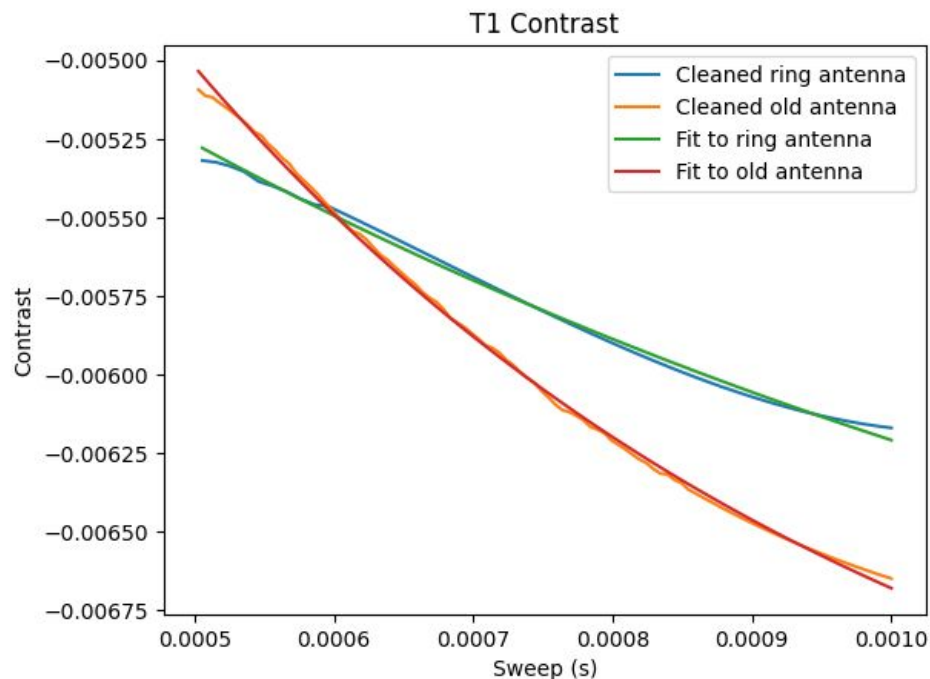
Rabi oscillations show coherent control capabilities.

- τ_{π} for old antenna: $0.190 \mu\text{s}$
- τ_{π} for single ring antenna: $0.141 \mu\text{s}$
- τ_{π} for omega antenna: $0.137 \mu\text{s}$
- Dephasing time for old antenna very short; basically impossible to fit
- Dephasing time for omega antenna: $\sim 0.24 \text{ ms}$
- Dephasing time for single ring antenna: $\sim 0.21 \text{ ms}$



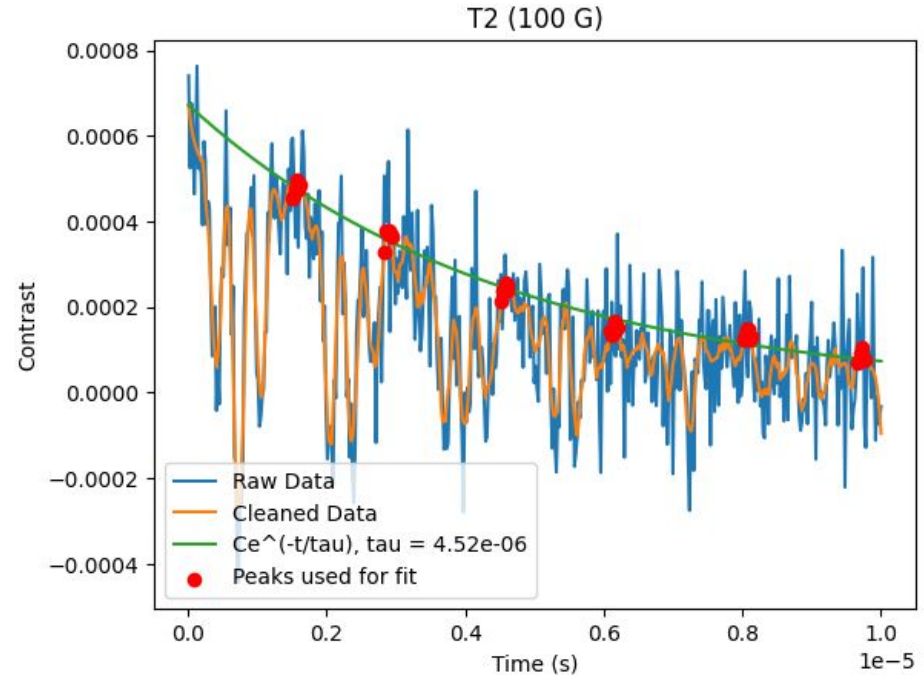
Successful T_1 achieved on single ring, better control.

- T_1 : measurement of thermal relaxation of spins
- Ring antenna run done with fewer data points to average over (10 vs 20), higher magnetic field (50 G vs 25 G)
- T_1 for old antenna: $518 \pm 14 \mu\text{s}$
- T_1 for new antenna: $996 \pm 132 \mu\text{s}$



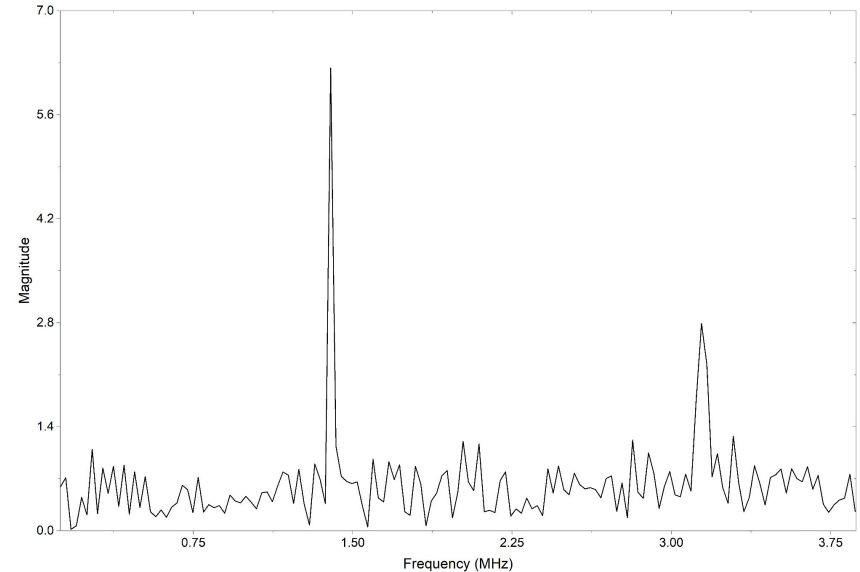
Successful T_2 achieved on single ring

- T_2 : measurement of relaxation of spin-spin coupling
- Find T_2 of $4.52 \pm 0.14 \mu\text{s}$
- See hyperfine splitting in triple peaks



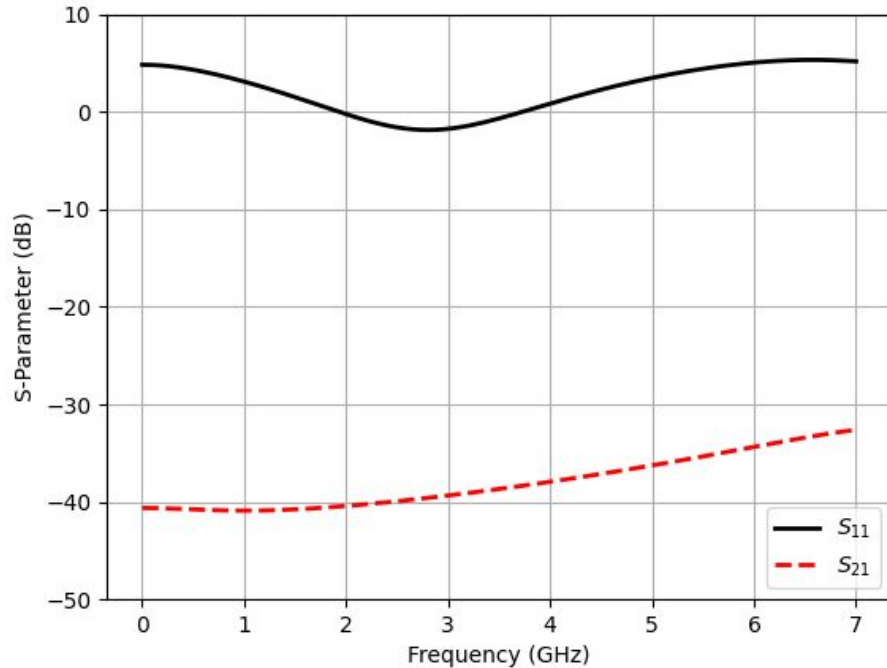
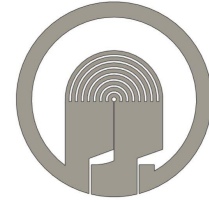
Conclusion

- Simulations match characterization well
- Omega and concentric antennas best antennas in characterization, ESR, Rabi
- Achieved T_1 , T_2 measurements on single ring with better control than old antenna
- Successfully designed and manufactured antennas which fit on a small cover slip
- Concentric completely new design that is more broadband/versatile
- Have already used to successfully obtain NMR spectra



First ^1H -NMR spectrum
from microscope

Future directions: concentric-omega antenna



- Simulation results uncertain - S_{11} does not make sense
- S_{21} absorption high, seems broadband which is promising
- Being manufactured right now, waiting on experimental data

References

- [1] Bucher, D. B., et al. (2019). "Quantum diamond spectrometer for nanoscale NMR and ESR spectroscopy." *Nature Protocols* 14(9): 2707-2747.
- [2] Kento Sasaki, Yasuaki Monnai, Soya Saijo, Ryushiro Fujita, Hideyuki Watanabe, Junko Ishi-Hayase, Kohei M. Itoh, Eisuke Abe; Broadband, large-area microwave antenna for optically detected magnetic resonance of nitrogen-vacancy centers in diamond. *Rev. Sci. Instrum.* 1 May 2016; 87 (5): 053904. <https://doi.org/10.1063/1.4952418>
- [3] Opaluch, O.R., Oshnik, N., Nelz, R., Neu, E. (2021). "Optimized Planar Microwave Antenna for Nitrogen Vacancy Center Based Sensing Applications." *Nanomaterials*, 11, 2108-2121. <https://doi.org/10.3390/nano11082108>
- [4] Davoodi, H., Nordin, N., Munakata, H. et al. Untuned broadband spiral micro-coils achieve sensitive multi-nuclear NMR TX/RX from microfluidic samples. *Sci Rep* 11, 7798(2021). <https://doi.org/10.1038/s41598-021-87247-2>

Acknowledgements

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Marvell Nanofabrication
Laboratory at UC Berkeley

Questions?

Nitrogen Vacancy (NV) Centers

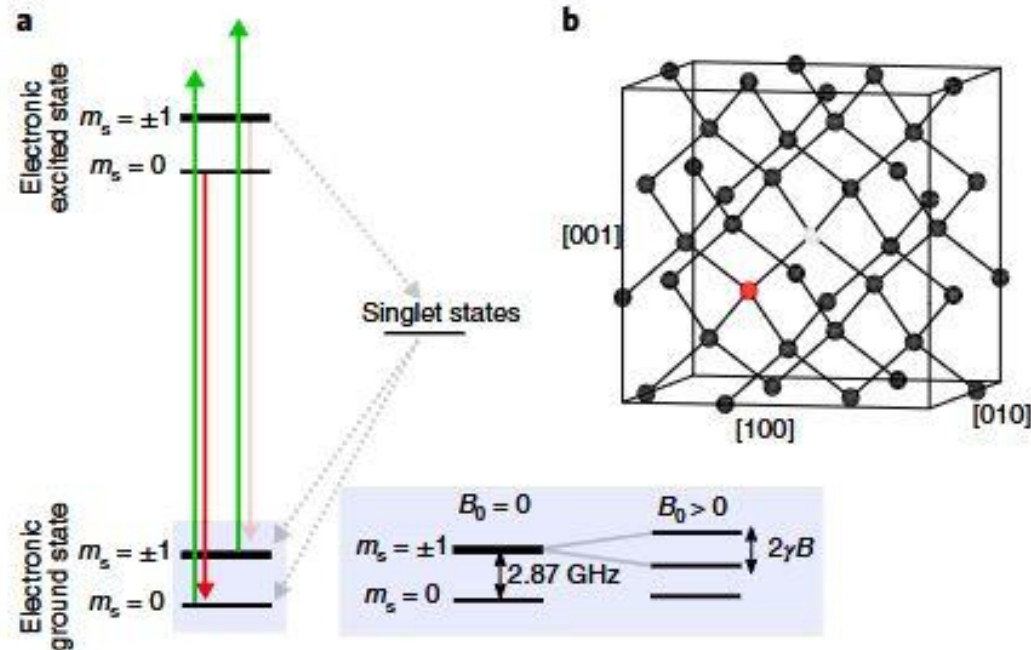


Figure from [1].

Controlling NV Centers

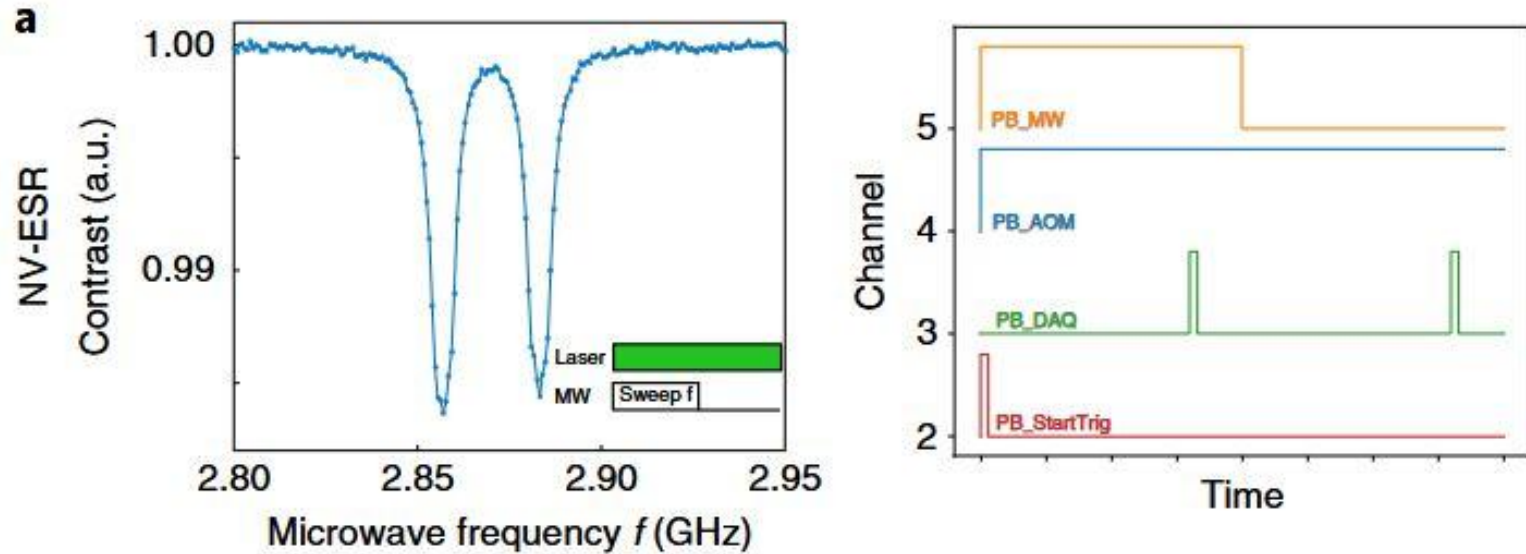
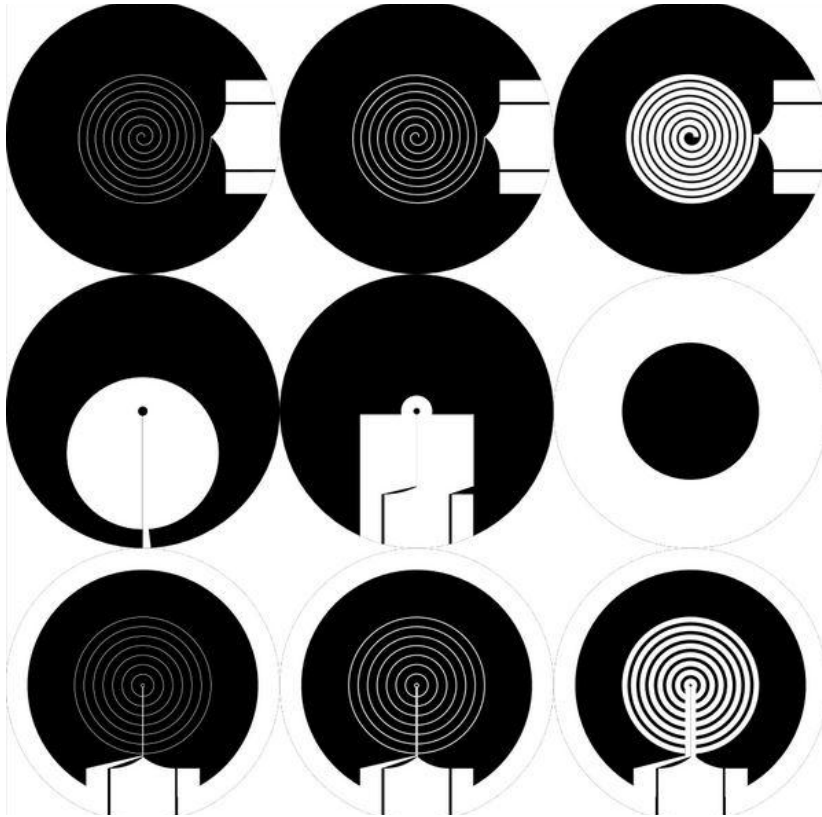


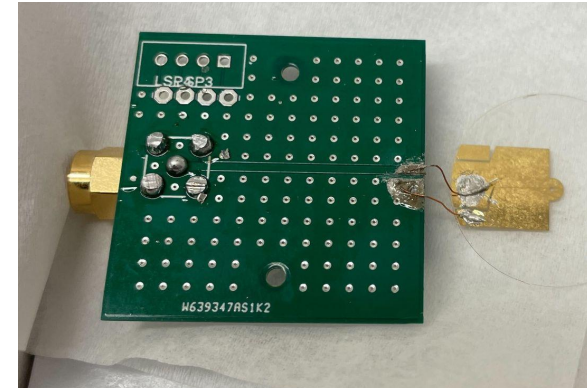
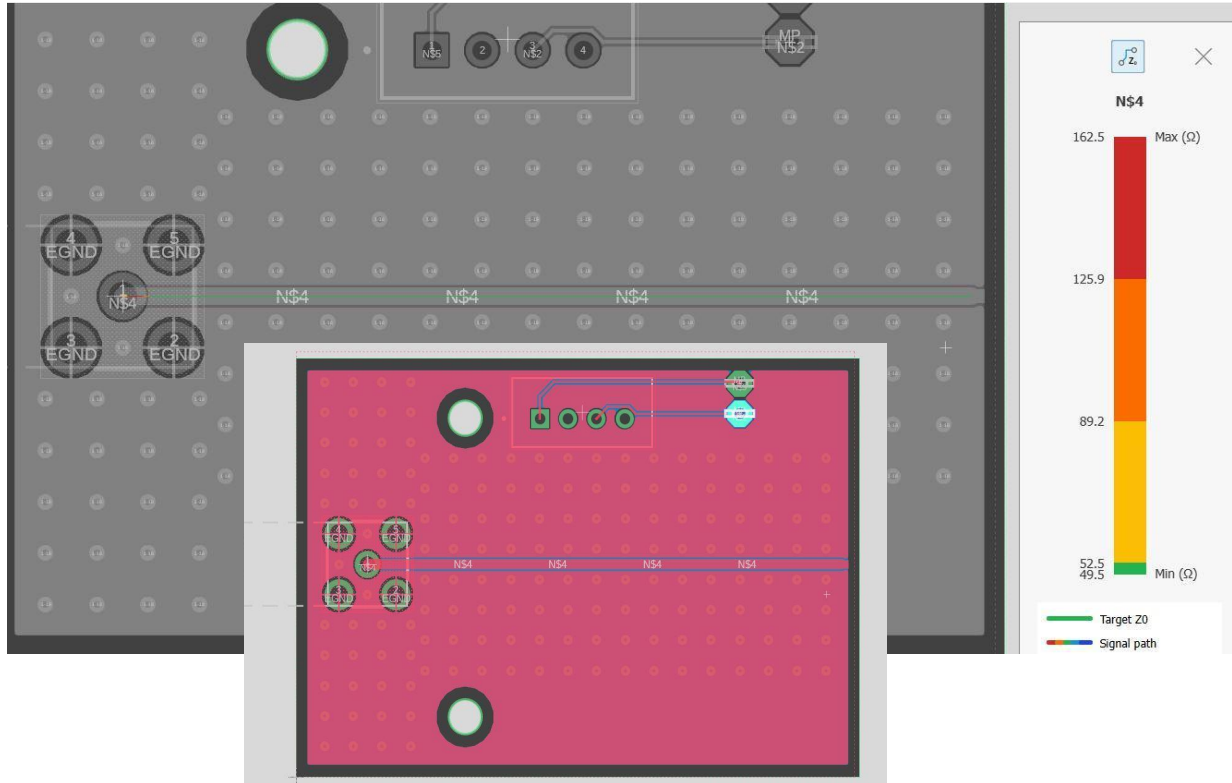
Figure from [1].

Manufacture



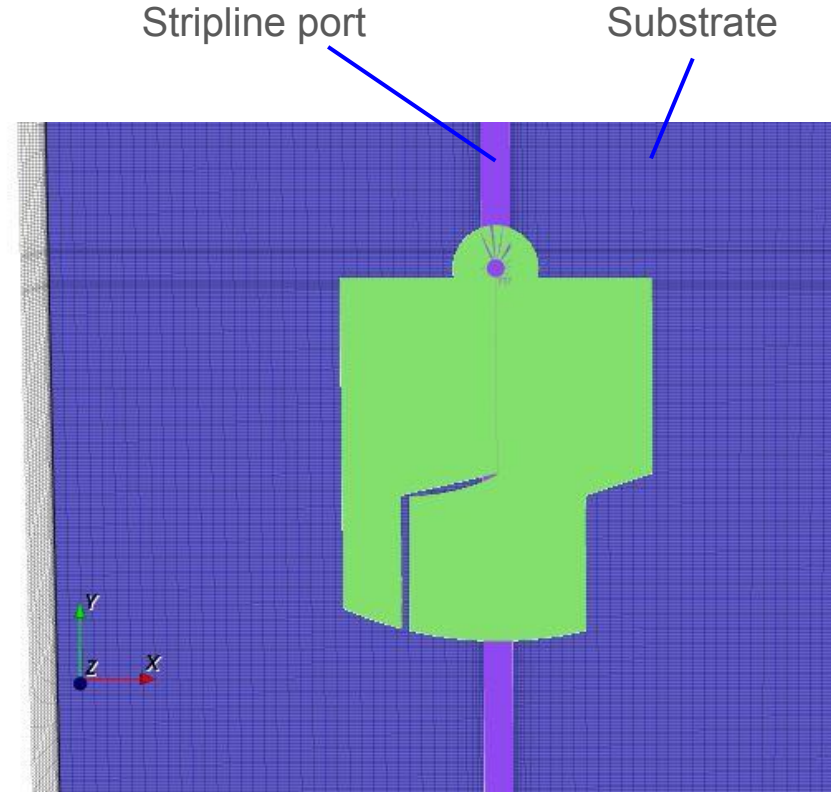
- Mask for antenna manufacture - white is gold, black is glass cover slip/no gold
- Top row: spiral antennas based on [4], spiral width $50\ \mu\text{m}$ to $500\ \mu\text{m}$
- Middle row: single ring antenna based on [2], omega antenna based on [3], ground ring
- Bottom row: new design, concentric antennas with ring width from $50\ \mu\text{m}$ to $500\ \mu\text{m}$
- Manufactured at Molecular Foundry, Marvell NanoLab
- 1" diameter cover slip, $100\ \mu\text{m}$ thick
- Gold $\sim 7.5\ \mu\text{m}$ thick

Microwave delivery

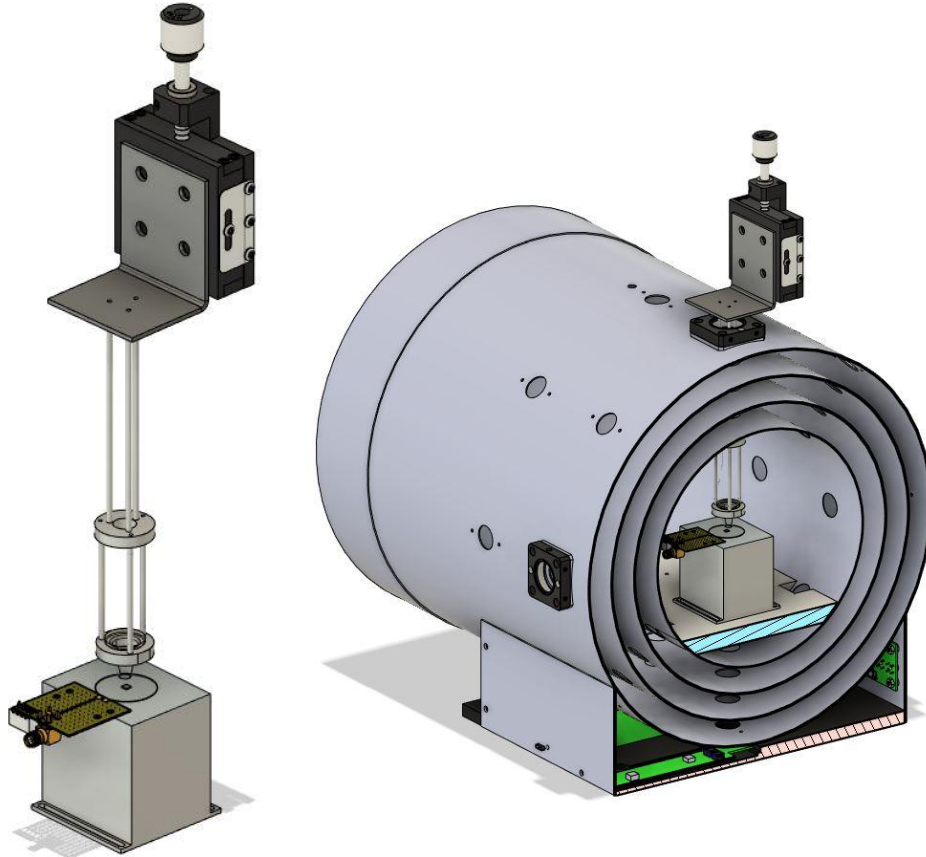


Simulation setup

- Used OpenEMS, open source Python package
- Assumed perfect electrical conductor (PEC) stripline ports, some dispersion affects which account for shifts
- Boundary conditions in six directions: Mur absorbing boundary condition (MUR), MUR, perfectly matched layer (PML), PML, PEC, MUR



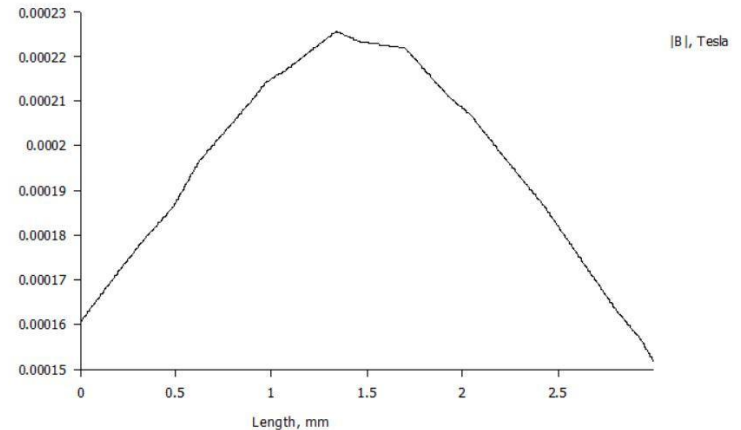
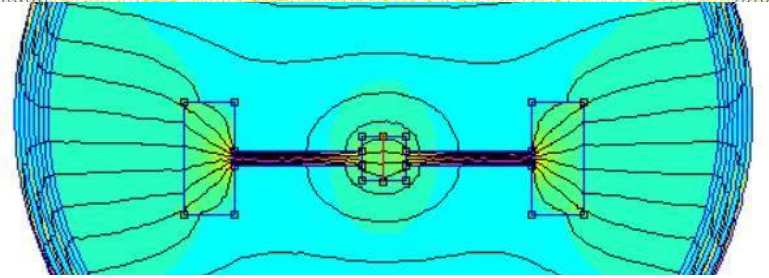
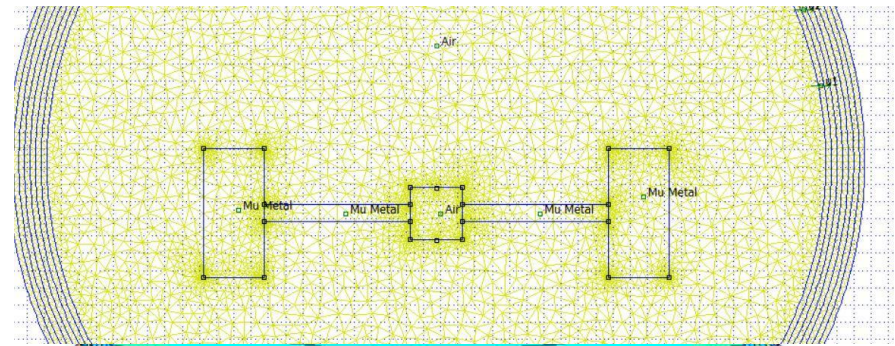
Magnetometer design



- Magnetic field shielding at right; most of mounting scheme to fit in this
- Parabolic concentrator to collect as much fluorescence as possible
- Antenna design to optimize delivery of microwave pulses
- Signal sensor head and reference sensor head to subtract out background
- Insulation to reduce thermal noise from laser

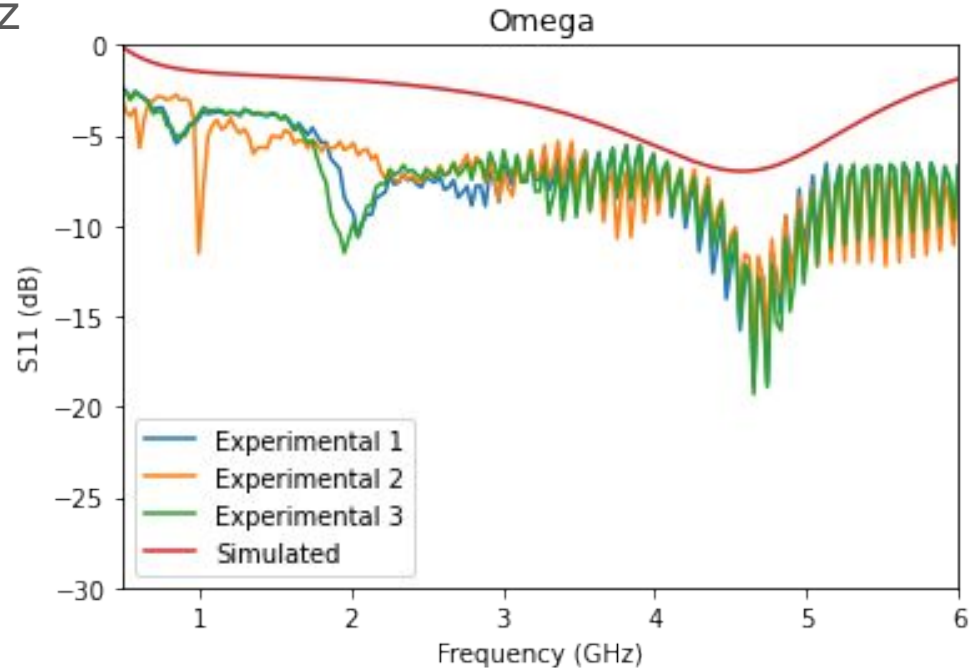
Flux concentrators

- Simulation done with constant magnetic field of $6.3 \times 10^{-5} \text{ T}$ (Earth's is $5 \times 10^{-5} \text{ T}$) for convenience purposes
- T design based on literature
- Did grid search sweeping width and length of bar and end cap (four parameters)
- Found dimensions for which highest flux concentration occurred: got field in center that was $\sim 3.7\times$ larger, doing further refinement



Single ring antennas

side by side comparison of 1.5-3.5 GHz
for concentric and omega, maybe
sasaki



Multi-ring antennas

Feedback

- need to add data on microwave delivery homogeneity
- need to add data on magnetic field homogeneity
- make note of spot size for antenna microwave delivery
- daipeng yang's talk?
- capacitor in antenna as noise filter
- peltier effect for cooling?
- collabs - 2-3 emails